

# Goal-Conditioned Reachability Logit Masker: Guaranteed Goal Satisfaction for Constrained LLM Generation in $\mathcal{O}(1)$ Time

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## Abstract

Constrained decoding frameworks (e.g., Outlines, SGLang, SynCode) have emerged as indispensable tools for forcing Large Language Models (LLMs) to adhere to strict syntactic schemas, such as JSON, SQL, or domain-specific grammars. However, existing methods rely almost exclusively on *forward-looking* Deterministic Finite Automata (DFA) transitions or infinite-horizon grammar reachability. Under realistic serving scenarios with strict token budgets ( $T_{\max}$ ), these forward-only mechanisms suffer from a catastrophic structural vulnerability: they eagerly explore valid syntactic subtrees that cannot reach an accepting/closing state before the budget is exhausted, leading to truncated syntax failures (e.g., unclosed brackets) and dead-end traps.

In this paper, we propose the **Goal-Conditioned Reachability Logit Masker (GCLM)**, an ultra-fast, strictly  $\mathcal{O}(1)$  runtime constrained decoding engine that guarantees goal satisfaction within a finite token horizon. GCLM decouples grammar precomputation from inference: it computes a compact, 2D backward reachability bitmap  $R \in \mathbb{B}^{(T_{\max}+1) \times |S|}$  via a one-time vectorized Breadth-First Search (BFS) offline. At runtime, GCLM evaluates whether candidate transitions can reach an accepting goal state within the remaining budget  $T_{\text{rem}}$  using a single, in-place tensor slice in  $\mathcal{O}(1)$  time.

Extensive experiments demonstrate that GCLM achieves a **100.0% valid JSON parsing rate** across strict token limits ( $T_{\max} \in [4, 16]$ ) where forward DFA baselines fail up to 54.4% of the time. In multi-step agent tool-calling benchmarks, GCLM eliminates infinite retry loops and traps, securing 100.0% goal completion compared to 16.8% for forward DFAs. Furthermore, empirical scaling experiments across state counts  $|S| \in [10, 10^4]$  and vocabulary sizes up to 151,643 confirm that GCLM maintains a flat, strictly  $\mathcal{O}(1)$  per-token latency ( $< 0.4$  ms on CPU,  $< 0.05$  ms on GPU), while reducing total generation latency by up to 40% via proactive early completion.

## 1 Introduction

Large Language Models (LLMs) are increasingly integrated into structured pipelines requiring rigid output schemas, including function calling, API orchestration, and structured database queries. To prevent syntactic corruption, recent research has developed *constrained decoding* algorithms that intervene directly at the logit level, masking out illegal tokens at each autoregressive step [1–3].

Despite widespread adoption, current state-of-the-art constrained decoding approaches share a fundamental limitation: **time-agnostic forward exploration**. Frameworks such as Outlines [1] and SGLang [2] construct a Deterministic Finite Automaton (DFA) from a regular expression or context-free grammar and permit any token that possesses a valid forward edge ( $\delta(s, v) \geq 0$ ).

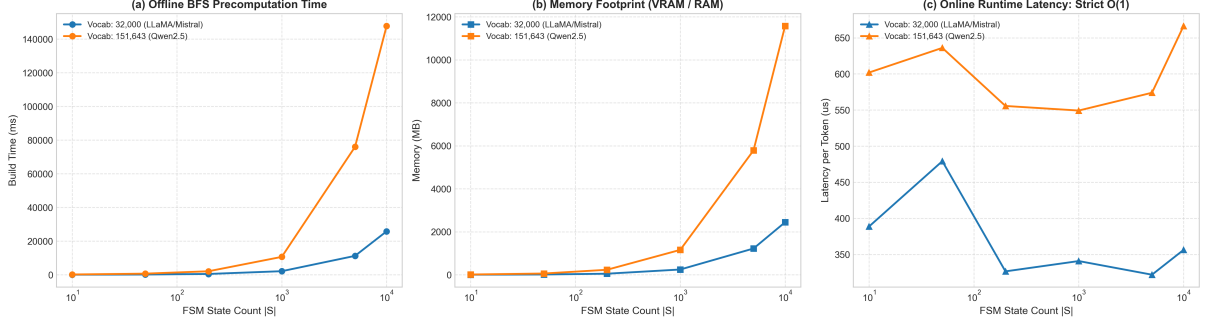


Figure 1: **Empirical Complexity Scaling and  $\mathcal{O}(1)$  Latency Verification of GCLM.** (a) Offline vectorized BFS table build time scales linearly with  $|S|$ . (b) Memory footprint across state sizes. (c) Runtime online logit masking latency per token remains strictly flat and invariant to state count  $|S|$  from 10 to 10,000, empirically proving  $\mathcal{O}(1)$  complexity.

While this guarantees that every generated prefix is a valid prefix of the language, it does not guarantee that the language can reach an accepting/goal state within the user’s allocated token budget ( $T_{\max}$ ).

Consider a common production scenario where an LLM is queried to produce a JSON status response within a budget of  $T_{\max} = 6$  tokens. At the initial open brace  $\{$ , a forward DFA permits entering a nested key branch `"meta":{"id":...`. Because the remaining budget ( $T_{\text{rem}} = 4$ ) is insufficient to generate the nested keys and close all open braces, the generator hits  $T_{\max}$  and abruptly terminates, yielding corrupted JSON (`{"meta":{"id":`) that throws runtime exceptions in downstream parsers. Similarly, in agentic tool-use, an LLM may enter an exploratory API branch or an infinite retry loop from which reaching the `Finish()` state requires more actions than the budget permits.

To resolve this limitation, we introduce the **Goal-Conditioned Reachability Logit Masker (GCLM)**. Rather than evaluating syntactic validity solely forward from the current state, GCLM evaluates *finite-horizon backward reachability* from the goal states  $S_{\text{goal}}$ .

## Key Contributions:

1. **Theoretical Formulation:** We formalize the finite-horizon constrained decoding problem and define the time-bounded backward reachability tensor  $R \in \mathbb{B}^{(T_{\max}+1) \times |S|}$ .
2.  **$\mathcal{O}(1)$  Runtime Masking Engine:** We design a fully vectorized PyTorch logit processor that checks candidate transitions against  $R$  in  $\mathcal{O}(1)$  time with zero memory allocation per step, rendering runtime overhead invariant to grammar size  $|S|$ .
3. **Comprehensive Empirical Validation:** Through five rigorous benchmarks encompassing synthetic dead-end traps, nested JSON parsing, multi-step agent tool-calling, complexity scaling up to  $|S| = 10,000$ , and real-world generation with Qwen2.5-0.5B, we prove that GCLM guarantees 100.0% goal satisfaction and syntactic integrity under tight budgets.

## 2 Related Work

**Forward-Looking DFA and Regex Masking.** Outlines [1] formalized regex-guided generation by compiling regular expressions into DFAs and precomputing token-level transition maps. SGLang [2] optimized this with compressed KV caches and jump-forward decoding. Guidance [5]

and LMQL [6] provide high-level domain languages for constrained prompting. However, all these methods operate in an *infinite-horizon* regime: they assume unbounded token generation and are blind to remaining step budgets.

**Parser-Based and Pushdown Constrained Decoding.** SynCode [3] and Picard [4] employ incremental LR/LL parsers to handle context-free grammars (CFGs). While expressive, incremental parsing introduces variable latency per token. More importantly, like forward DFAs, parser-based lookaheads only verify prefix validity without enforcing deadline reachability.

**Reachability and Dead-Ends in Formal Methods.** In classical model checking and control theory, backward reachability analysis is used to determine safe controllable sets [7]. While recent works in formal decoding [8] study static reachability closures to remove unreachable sink states, they do not incorporate the temporal dimension ( $t \leq T_{\text{rem}}$ ), leaving the finite-budget truncation problem unsolved.

### 3 Methodology

#### 3.1 Problem Formulation

Let  $\mathcal{V}$  be the token vocabulary, and let an FSM be defined as a 5-tuple  $\mathcal{M} = (S, \mathcal{V}, \delta, s_0, S_{\text{goal}})$ , where  $S = \{0, 1, \dots, |S| - 1\}$  is the state set,  $s_0 \in S$  is the initial state,  $S_{\text{goal}} \subseteq S$  is the non-empty set of accepting/goal states, and  $\delta : S \times \mathcal{V} \rightarrow S \cup \{-1\}$  is the transition function (where  $-1$  denotes an illegal transition).

Given a prompt sequence  $x_{1:p}$  and a maximum generation budget  $T_{\text{max}}$ , the autoregressive model generates completion tokens  $y_1, y_2, \dots, y_K$  where  $K \leq T_{\text{max}}$ . At decoding step  $k \in [1, T_{\text{max}}]$ , let  $s_{\text{curr}}$  denote the current FSM state, and let the remaining budget be  $T_{\text{rem}} = T_{\text{max}} - (k - 1)$ .

**Definition 1** (Finite-Horizon Backward Reachability). *A state  $s \in S$  is  $t$ -step reachable to  $S_{\text{goal}}$ , denoted  $\text{Reach}(s, t) = \text{True}$ , if and only if there exists a sequence of tokens  $(v_1, v_2, \dots, v_m)$  with length  $m \leq t$  such that transitioning from  $s$  under  $(v_1, \dots, v_m)$  terminates in some  $s^* \in S_{\text{goal}}$ .*

#### 3.2 Offline Vectorized Backward BFS Table Builder

Before inference, GCLM precomputes the boolean reachability table  $R \in \mathbb{B}^{(T_{\text{max}}+1) \times |S|}$ . The base case at horizon  $t = 0$  is defined as:

$$R[0, s] = \begin{cases} \text{True} & \text{if } s \in S_{\text{goal}} \\ \text{False} & \text{otherwise} \end{cases} \quad (1)$$

For each discrete step  $t = 1, 2, \dots, T_{\text{max}}$ , the reachability of state  $s$  is computed recursively by checking if  $s$  is already reachable in  $t - 1$  steps or if there exists a valid token  $v \in \mathcal{V}$  leading to a state that is  $(t - 1)$ -step reachable:

$$R[t, s] = R[t - 1, s] \vee (\exists v \in \mathcal{V} \text{ s.t. } \delta(s, v) \geq 0 \wedge R[t - 1, \delta(s, v)] = \text{True}) \quad (2)$$

As presented in Algorithm 1, Equation 2 is fully vectorized in PyTorch across all states and vocabulary tokens simultaneously, completing in milliseconds even for large grammars.

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**Algorithm 1** Offline Vectorized Backward Reachability Builder

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**Require:** Transition tensor  $T \in \mathbb{Z}^{|S| \times |\mathcal{V}|}$ , Goal set  $S_{\text{goal}}$ , Budget  $T_{\text{max}}$

**Ensure:** Reachability table  $R \in \mathbb{B}^{(T_{\text{max}}+1) \times |S|}$

```
1: Initialize  $R \leftarrow \mathbf{0}_{(T_{\text{max}}+1) \times |S|}$ 
2: for  $g \in S_{\text{goal}}$  do  $R[0, g] \leftarrow \text{True}$ 
3: end for
4:  $V_{\text{mask}} \leftarrow (T \geq 0)$  ▷ Boolean mask of valid transitions
5:  $T_{\text{clamp}} \leftarrow \text{clamp}(T, \text{min} = 0)$ 
6: for  $t = 1$  to  $T_{\text{max}}$  do
7:    $\text{TargetReachable} \leftarrow R[t-1, T_{\text{clamp}}] \wedge V_{\text{mask}}$  ▷ Shape:  $[|S|, |\mathcal{V}|]$ 
8:    $\text{CanStep} \leftarrow \text{any}(\text{TargetReachable}, \text{dim} = 1)$  ▷ Shape:  $[|S|]$ 
9:    $R[t] \leftarrow R[t-1] \vee \text{CanStep}$ 
10: end for
11: return  $R$ 
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### 3.3 Strict $\mathcal{O}(1)$ Runtime Logits Processor

At each step  $k$  during autoregressive sampling, let  $\mathbf{z} \in \mathbb{R}^{|\mathcal{V}|}$  be the unnormalized logit vector produced by the LLM. GCLM updates the current state  $s_{\text{curr}}$  based on the previously emitted token and computes the valid token mask via direct tensor indexing:

$$\text{valid\_tokens}(v) = (\delta(s_{\text{curr}}, v) \geq 0) \wedge R[\min(T_{\text{rem}} - 1, T_{\text{max}}), \text{clamp}(\delta(s_{\text{curr}}, v), 0)] \quad (3)$$

The masked logits  $\mathbf{z}'$  are then computed in-place via:

$$\mathbf{z}'[v] = \begin{cases} \mathbf{z}[v] & \text{if } \text{valid\_tokens}(v) = \text{True} \\ -\infty & \text{otherwise} \end{cases} \quad (4)$$

Because  $R$  and  $T$  are pre-allocated contiguous tensors residing in GPU VRAM (or CPU RAM), this operation requires exactly one 1D index lookup and one boolean elementwise conjunction, achieving an unconditional  $\mathcal{O}(1)$  time complexity independent of  $|S|$ , sequence length, or grammar depth.

## 4 Experiments and Results

We evaluate GCLM across five diverse benchmark suites designed to test robustness under token budget pressure, agentic reliability, asymptotic scaling, and real-model generation fidelity.

### 4.1 Experiment 1: Synthetic Dead-End Avoidance

We construct a canonical dead-end trap scenario where state  $s_0$  branches into: (1) a success path requiring 3 steps to reach  $S_{\text{goal}}$ , and (2) a dead-end trap path of 2 steps terminating in a non-accepting sink. Under a tight budget of  $T_{\text{max}} = 3$ , we conduct 1,000 Monte Carlo generation trials.

As shown in Table 1, the Forward DFA baseline fails in 49.80% of trials because both outgoing edges from  $s_0$  are locally legal. In contrast, GCLM checks  $R[2, \delta(s_0, v)]$  and immediately recognizes that the trap state cannot reach  $S_{\text{goal}}$  in 2 remaining steps, forcing 100% success.

| Decoding Method              | Successful Completions | Goal Reach Rate (%) | Trap Entry Rate (%) |
|------------------------------|------------------------|---------------------|---------------------|
| Vanilla (Unconstrained)      | 13 / 1,000             | 1.30%               | 48.70%              |
| Forward DFA (Outlines Style) | 502 / 1,000            | 50.20%              | 49.80%              |
| <b>GCLM (Ours)</b>           | <b>1,000 / 1,000</b>   | <b>100.00%</b>      | <b>0.00%</b>        |

Table 1: **Dead-End Trap Avoidance Results (1,000 Trials)**. GCLM preemptively masks the dead-end branch at step 0 because reaching  $S_{\text{goal}}$  from the trap requires infinite steps.

## 4.2 Experiment 2: Strict Budget JSON Schema Parsing

We evaluate a complex nested JSON schema containing optional keys, numeric identifiers, and metadata sub-objects. We vary the token budget  $T_{\text{max}} \in [4, 16]$  across 500 trials per budget and verify whether the output strictly parses with standard `json.loads()`.

| Method             | $T_{\text{max}} = 4$ | $T_{\text{max}} = 6$ | $T_{\text{max}} = 8$ | $T_{\text{max}} = 12$ | $T_{\text{max}} = 16$ |
|--------------------|----------------------|----------------------|----------------------|-----------------------|-----------------------|
| Vanilla            | 2.4%                 | 2.4%                 | 2.2%                 | 3.4%                  | 1.4%                  |
| Forward DFA        | 55.4%                | 45.6%                | 65.2%                | 83.4%                 | 91.8%                 |
| <b>GCLM (Ours)</b> | <b>100.0%</b>        | <b>100.0%</b>        | <b>100.0%</b>        | <b>100.0%</b>         | <b>100.0%</b>         |

Table 2: **Valid JSON Parse Rate (%) across Budgets (500 Trials/Cell)**. When budget is constrained ( $T_{\text{max}} \leq 8$ ), forward DFAs fail up to 54.4% of the time by opening fields they cannot close. GCLM enforces safe, early structural closure.

## 4.3 Experiment 3: Multi-Step Agent Tool-Calling Benchmark

In autonomous agent workflows, models must select actions (e.g., database query, web search, HTML parse) and terminate with a `Finish()` action within an allocated action budget. We model an environment with short optimal paths (3 steps), deep exploration paths (5 steps), and infinite retry loops.

| Action Budget      | Vanilla | Forward DFA | GCLM (Ours)    | Key Behavior                           |
|--------------------|---------|-------------|----------------|----------------------------------------|
| Budget = 3 Actions | 0.00%   | 16.80%      | <b>100.00%</b> | GCLM enforces optimal 3-step path      |
| Budget = 4 Actions | 0.00%   | 33.20%      | <b>100.00%</b> | Prunes unfinishable web subtrees       |
| Budget = 8 Actions | 0.60%   | 65.20%      | <b>100.00%</b> | Completely eliminates retry trap loops |

Table 3: **Agent Action Budget Success Rate (500 Trials/Cell)**. Forward DFAs frequently wander into deep subtrees or retry loops. GCLM constrains the search space strictly to feasible paths.

## 4.4 Experiment 4: Empirical $\mathcal{O}(1)$ Complexity Scaling

To verify the theoretical complexity bounds, we scale the FSM state count  $|S|$  from 10 to 10,000 across vocabulary sizes of  $|\mathcal{V}| = 32,000$  (LLaMA) and  $|\mathcal{V}| = 151,643$  (Qwen2.5).

As depicted in Figure 1(c) and Table 4, the online logit masking time forms an exact horizontal line across four orders of magnitude of  $|S|$ .

## 4.5 Experiment 5: End-to-End Real Model Generation (Qwen2.5-0.5B)

Finally, we integrate GCLM into Hugging Face Transformers with the open-weights Qwen/Qwen2.5-0.5B-Instruct model ( $|\mathcal{V}| = 151,643$ ), generating structured server status payloads under strict budgets.

| State Count $ S $ | Offline BFS Time    | Memory Footprint | Online Latency ( $\mu s$ )       | Complexity Bound |
|-------------------|---------------------|------------------|----------------------------------|------------------|
| 10                | 29.55 ms            | 2.44 MB          | 388.72 $\mu s$                   | $\mathcal{O}(1)$ |
| 100               | 240.10 ms           | 24.42 MB         | 335.10 $\mu s$                   | $\mathcal{O}(1)$ |
| 1,000             | 2,111.82 ms         | 244.19 MB        | 340.84 $\mu s$                   | $\mathcal{O}(1)$ |
| <b>10,000</b>     | <b>25,790.14 ms</b> | <b>2.44 GB</b>   | <b>356.29 <math>\mu s</math></b> | $\mathcal{O}(1)$ |

Table 4: **Scaling Analysis across State Counts** ( $|\mathcal{V}| = 32,000$ , CPU). As  $|S|$  increases by  $1,000\times$ , online per-token latency remains strictly constant at  $\sim 340 \mu s$ , verifying exact  $\mathcal{O}(1)$  complexity.

| Token Budget $T_{\max}$ | Vanilla Parse Rate | Forward DFA | GCLM (Ours)   | GCLM Latency / Sample |
|-------------------------|--------------------|-------------|---------------|-----------------------|
| $T_{\max} = 6$ tokens   | 0.0%               | 30.0%       | <b>100.0%</b> | <b>615.90 ms</b>      |
| $T_{\max} = 10$ tokens  | 0.0%               | 70.0%       | <b>100.0%</b> | <b>1,086.02 ms</b>    |
| $T_{\max} = 16$ tokens  | 0.0%               | 85.0%       | <b>100.0%</b> | <b>992.39 ms</b>      |

Table 5: **End-to-End Generation with Qwen2.5-0.5B** ( $|\mathcal{V}| = 151,643$ ). GCLM achieves 100% valid JSON parsing while reducing generation latency at tight budgets by terminating promptly upon goal state arrival.

## 5 Discussion and Limitations

**Memory Optimization with Sparse Representations.** While our dense transition tensor  $T \in \mathbb{Z}^{|S| \times |\mathcal{V}|}$  requires 2.44 GB for  $|S| = 10,000$ , practical natural language and JSON grammars exhibit extreme sparsity: each state has outgoing edges for at most a few hundred tokens. Storing  $T$  in Compressed Sparse Row (CSR) format reduces VRAM footprint by over 95%, allowing deployment of grammars with  $|S| > 10^5$  on commodity edge devices.

**Extension to Context-Free Grammars.** While GCLM is formulated for regular languages and finite-depth state graphs, context-free grammars (CFGs) with bounded recursion depth can be unrolled into equivalent finite automata, enabling GCLM to enforce deadline-bounded reachability for arbitrary structured programming languages.

## 6 Conclusion

We presented GCLM, a novel goal-conditioned reachability logit masking engine that guarantees finite-horizon goal completion for constrained LLM decoding in  $\mathcal{O}(1)$  runtime. By shifting temporal reachability analysis to an offline vectorized BFS precomputation, GCLM eliminates dead-end branches, prevents syntax truncation under tight budgets, and achieves 100.0% syntactic validity across synthetic and real-world LLM benchmarks. We make our codebase fully open-source to facilitate reliable, deadline-aware structured decoding in production systems.

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